

Introduction to Special Relativity

Chapter 2: Time Dilation and Length Contraction

Study Notes Series

1 The Light Clock Thought Experiment

Imagine a simple clock made of two mirrors facing each other, a fixed distance d apart, with a single photon of light bouncing between them. Each round trip of the photon counts as one "tick."

For an observer at rest with respect to the clock, one tick takes

$$\Delta t_0 = \frac{2d}{c}. \quad (1)$$

Now suppose the same clock moves past a second observer at speed v . From this observer's point of view, the photon does not simply travel straight up and down; it must also move sideways along with the clock. The photon therefore traces a longer, diagonal path for each tick. Since the speed of light c is the same for every observer (the second postulate from Chapter 1), a longer path means a longer time per tick.

Working through the geometry gives the famous result:

$$\Delta t = \gamma \Delta t_0, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (2)$$

This is **time dilation**: a clock moving relative to an observer runs slower, as measured by that observer, by the factor γ . Crucially, this is not an illusion or a measurement error. Both observers are correct in their own frame; time itself elapses differently.

A real-world confirmation: muons

Muons are unstable particles created when cosmic rays strike the upper atmosphere, roughly 15 km above the ground. Muons decay with a mean lifetime of about $2.2 \mu\text{s}$ in their own rest frame. Even traveling near the speed of light, a naive calculation suggests they should travel only around 660 m before decaying, far short of reaching the ground. Yet muons are detected at sea level in large numbers. Time dilation explains the discrepancy: from Earth's frame, the muon's internal clock runs slow by the factor γ , so it survives long enough to complete the trip.

2 Length Contraction

A closely related effect concerns length. Suppose an object has length L_0 when measured at rest (its **proper length**). An observer watching that object fly past at speed v measures a shorter

length:

$$L = \frac{L_0}{\gamma}. \quad (3)$$

Length contraction only occurs along the direction of motion; dimensions perpendicular to the motion are unaffected. From the muon's own point of view, it is not that time runs slowly; instead, the distance through the atmosphere is contracted, so it only has to cross a much shorter distance before decaying. Both explanations, time dilation from Earth's frame and length contraction from the muon's frame, predict the same observable outcome: more muons reach the ground than Newtonian physics would allow.

3 Putting Numbers to It

At $v = 0.99c$, the Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1 - 0.99^2}} \approx 7.09. \quad (4)$$

A clock moving at this speed ticks about 7 times slower as seen from a stationary frame, and an object at this speed appears about 7 times shorter along its direction of travel.

Quick check

A spaceship of proper length $L_0 = 100$ m flies past Earth at $v = 0.8c$, where $\gamma = 5/3$. What length does an Earth observer measure? (Answer: $L = 100/(5/3) = 60$ m.)